

Weekly Assessment #5 (5B9 - C5)

Course: Data Analytics

Topic: Data Summarization

Assessment Type: Written Typed

Submission Deadline: November 14, 2025, 11:59 PM

Total Marks: 50

N.B: For PART A, B use google sheet and share the link in your Document file with PART C.

Dataset:

https://docs.google.com/spreadsheets/d/1TkEQYICVDYY7xyawe2IrvndCqJ5cw7n9xPAoRI8Y_CQ/edit?usp=sharing

Part A (30 Marks)

Task 1: Regional Sales Summary (8 Marks)

1. Find total Sale using SUM() - **Units_Sold*Unit_Price**, then Create a Pivot Table showing total yearly sales by region.
2. Identify: the top-performing and lowest-performing regions.
3. Describe any visible trend across years.
4. Add a Pivot Chart (line or column) to visualize the trend.

Task 2: Customer and Category Insights (6 Marks)

1. Use a Pivot Table or SUMIFS formula to summarize sales by customer type and product category.
2. Identify: the customer type that contributes most to total sales, and the most popular category for each customer type.

Task 3: Formula Application (12 Marks)

Apply and show the use of the following formulas (2 marks each):

1. SUMIFS() – Total sales for Laptops sold in 2024.
2. AVERAGEIFS() – Average sales for Wholesale customers.
3. COUNTIFS() – Number of transactions in the East region.
4. IF() – Label sales as 'High' if Total_Sales > 30000, otherwise 'Low'.
5. VLOOKUP() – Display Product Category when given a Product Name.
6. XLOOKUP() – find the **Total Sales** of Laptops

For each: write the formula and explain what the result represents.

Task 4: Product Ranking (4 Marks)

1. Create a summary of total sales by product name.
2. Rank products from highest to lowest total sales.
 - Write a short observation (2–3 lines) on which products are driving overall revenue.

Part B (10 Marks)

Task 1: Regional Strategy (10 Marks)

1. Identify the region with the lowest sales.
2. Suggest two data-backed strategies to improve its performance.
 - Justify using summarized results.

Part C (10 Marks)

Answer each in 8-10 lines:

1. Define Data Summarization and explain why it is crucial in business analytics. (3 Marks)
2. Discuss two advantages and one limitation of Excel summarization tools. (3 Marks)
3. Explain how data summarization helps managers make data-driven decisions. (4 Marks)